

09/09/2026

CHAPTER 1: DIFFERENTIAL OPERATORS

1.1 Recalls, notation

- $\mathbb{N}$: natural numbers $\{0, 1, 2, \dots\}$ C/C++ unsigned int
- $\mathbb{Z}$: integer numbers $\{\dots, -2, -1, 0, 1, 2, \dots\}$ int
- $\mathbb{R}$: real float, double
32 bit, 64 bit
- $\mathbb{C}$: complex
- Tuples: for $n=2$ $(x_1, x_2) = (x, y) \in \mathbb{R}^2$
for $n=3$ $(x_1, x_2, x_3) = (x, y, z) \in \mathbb{R}^3$
for $n \in \mathbb{N}$ $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$
 $n > 0$

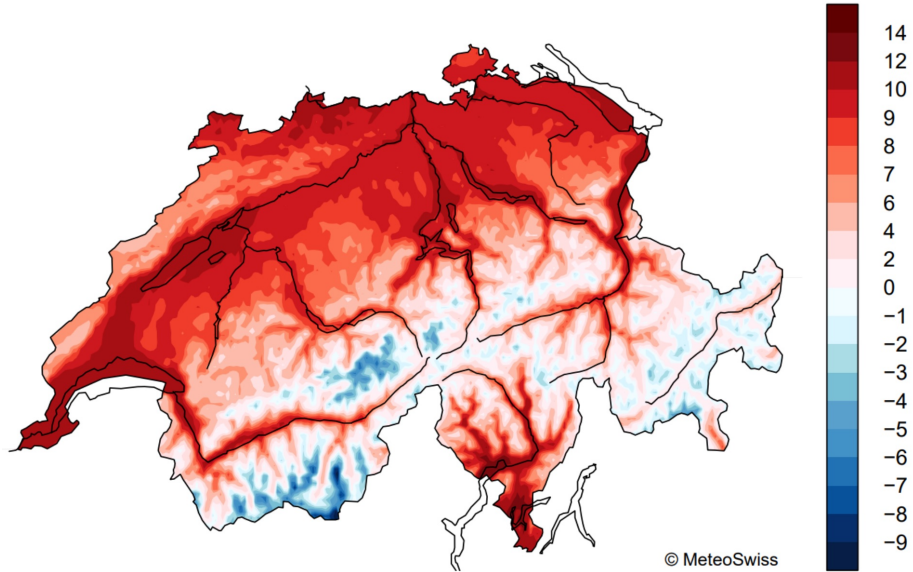
- A function that has real values defined on $\Omega \subset \mathbb{R}^n$ ($n > 0$)

$$f: \Omega \subset \mathbb{R}^n \longrightarrow \mathbb{R}$$

$$x \longmapsto f(x) = f(x_1, x_2, \dots, x_n)$$

$\vec{x}$
a tuple

is a **scalar field** defined on Ω



- For $k \in \mathbb{N}$ we write that $f \in C^k(\Omega)$ if all the derivatives (respect to $x_1, x_2, \dots, x_n$) with order $\leq k$ exist and are continuous.

- A function that has vector values and is defined on $\Omega \subset \mathbb{R}^n$

$$F: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$x \mapsto F(x) = (F_1(x), F_2(x), \dots, F_m(x))$$

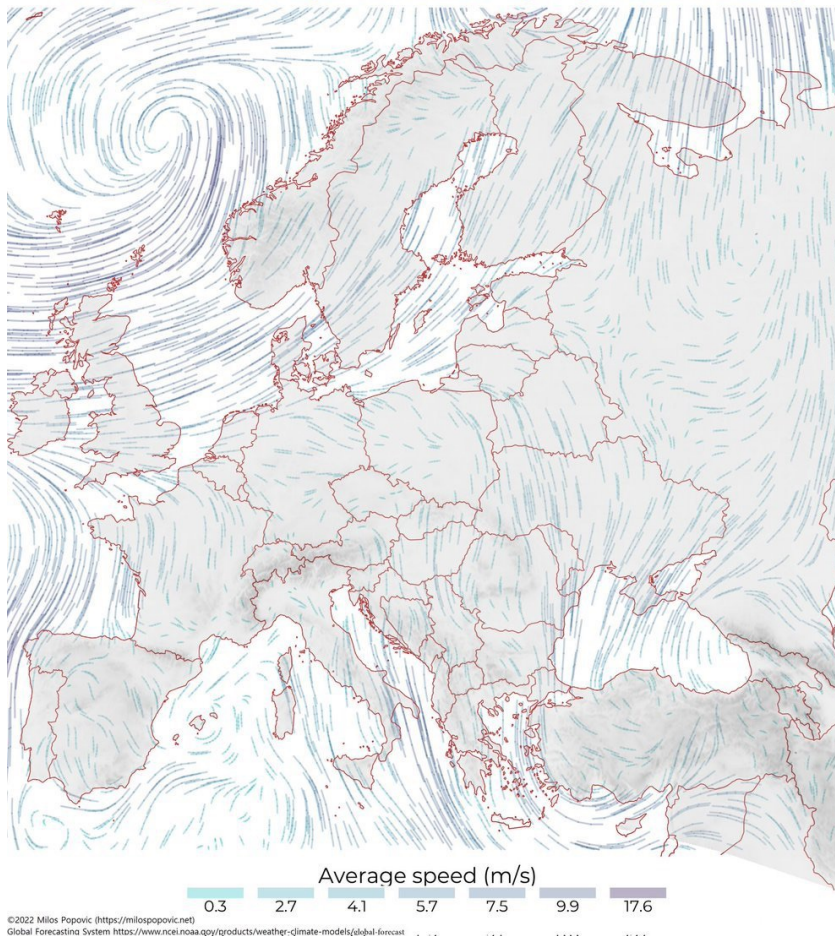
$\nearrow$
 tuple

where $F_i: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$

$$x \mapsto F_i(x) = F_i(x_1, \dots, x_n)$$

is a vector field.

Average wind speed in Europe (12 June 2022)



- For $k \in \mathbb{N}$ we write $F \in C^k(\Omega, \mathbb{R}^m)$ if $F_i \in C^k(\Omega)$ for $i=1, \dots, m$

- The differential operator **nabla**, denoted as ∇ , is defined as

$$\nabla = \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \dots, \frac{\partial}{\partial x_n} \right)$$

1.2 The gradient operator

1.2.1 Definition

Let be $\Omega \subset \mathbb{R}^n$ an open domain and

$$\begin{aligned} f: \Omega &\rightarrow \mathbb{R} \\ x &\mapsto f(x) = f(x_1, \dots, x_n) \end{aligned}$$

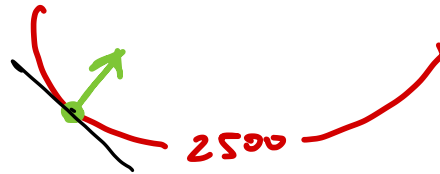
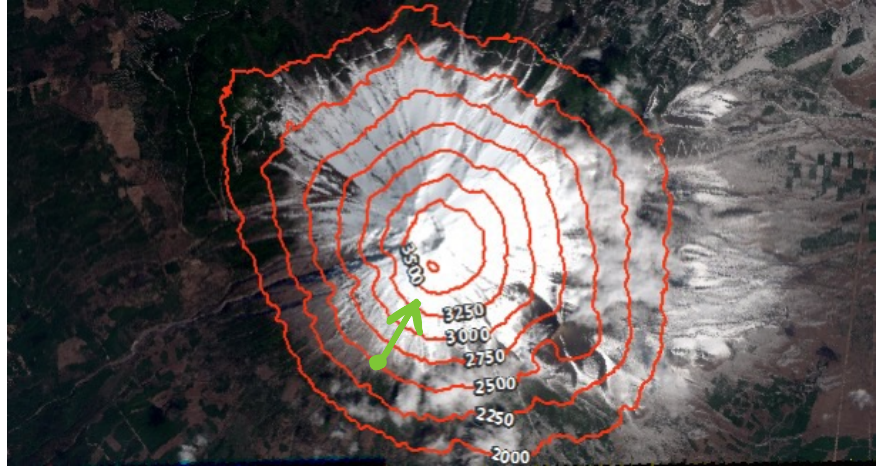
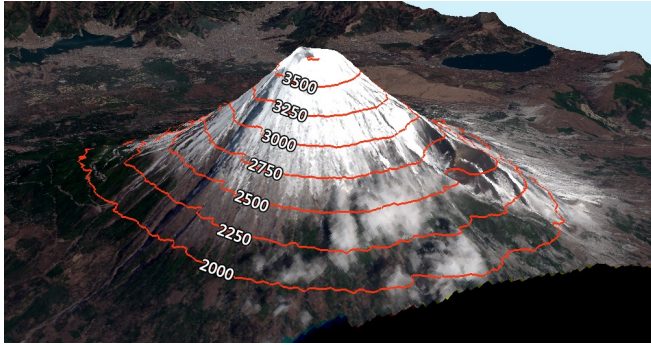
$$\text{s.t. } f \in C^1(\Omega)$$

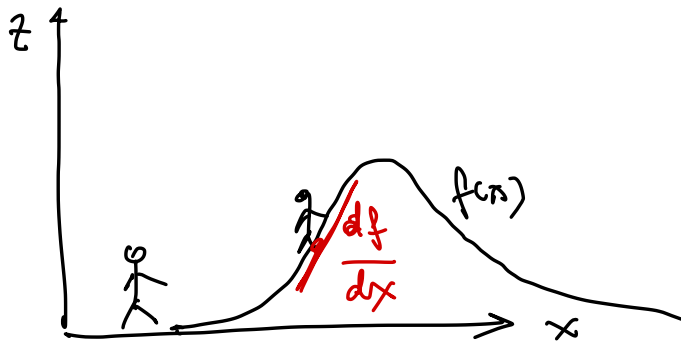
The gradient of f is denoted as $\text{grad } f$

(or ∇f , or Df) and is defined as

$$\text{grad } f(x) = \left(\frac{\partial f}{\partial x_1}(x), \frac{\partial f}{\partial x_2}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right) \in \mathbb{R}^n$$

$$\begin{aligned} \nabla f &= \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right) f \end{aligned}$$





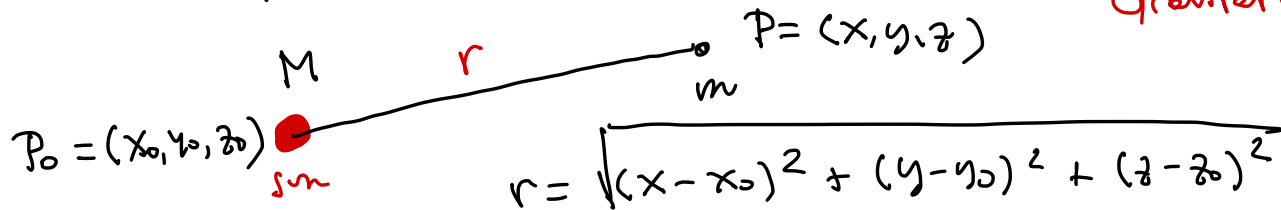
1.2.2 Examples:

• Example 1: let be $f: \mathbb{R}^3 \rightarrow \mathbb{R}$

$$(x, y, z) \mapsto f(x, y, z) = x^2 y^3 \sin(z^2)$$

$$\text{grad } f(x, y, z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \left(2xy^3 \sin(z^2), 3x^2 y^2 \sin(z^2), 2x^2 y^3 z \cos(z^2) \right) \in \mathbb{R}^3$$

• Example 2:



Gravitational field

$$f: \mathbb{R}^3 \setminus P_0 \rightarrow \mathbb{R}$$

$$(x, y, z) \mapsto f(x, y, z) = \frac{g m M}{r(x, y, z)}$$

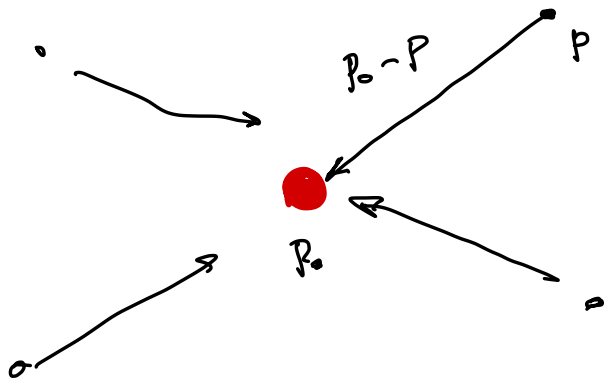
$$f(x, y, z) = \psi(r) = \psi(r(x, y, z)) = \frac{c}{r(x, y, z)}, \quad c = g m M.$$

$$\text{grad } f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$\frac{\partial f}{\partial x} = \frac{d\psi}{dr} \frac{\partial r}{\partial x}, \quad \frac{d\psi}{dr} = \frac{-c}{r^2}, \quad \frac{\partial r}{\partial x} = \frac{x - x_0}{r}$$

$$\frac{\partial f}{\partial x} = -\frac{c}{r^3} (x-x_0), \quad \frac{\partial f}{\partial y} = \dots, \quad \frac{\partial f}{\partial z} = \dots$$

$$\text{grad } f(x, y, z) = -\frac{c}{r^3} \underbrace{(x-x_0, y-y_0, z-z_0)}_{P(x, y, z) - P_0}$$



gravitational force

1.3 The divergence operator

let be $\Omega \subset \mathbb{R}^n$ on open domain and

$$F: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$\text{s.t. } F \in C^1(\Omega, \mathbb{R}^n)$$

The divergence operator $\nabla \cdot$ applied to F , denoted as $\text{div } F$ ($\nabla \cdot F, \div F$)

is defined as

$$\text{div } F(x) = \frac{\partial F_1}{\partial x_1}(x) + \frac{\partial F_2}{\partial x_2}(x) + \dots + \frac{\partial F_n}{\partial x_n}(x) \in \mathbb{R}$$

$$\text{Note: } \nabla \cdot F = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right) \cdot (F_1, \dots, F_n) = \frac{\partial F_1}{\partial x_1} + \dots + \frac{\partial F_n}{\partial x_n}$$